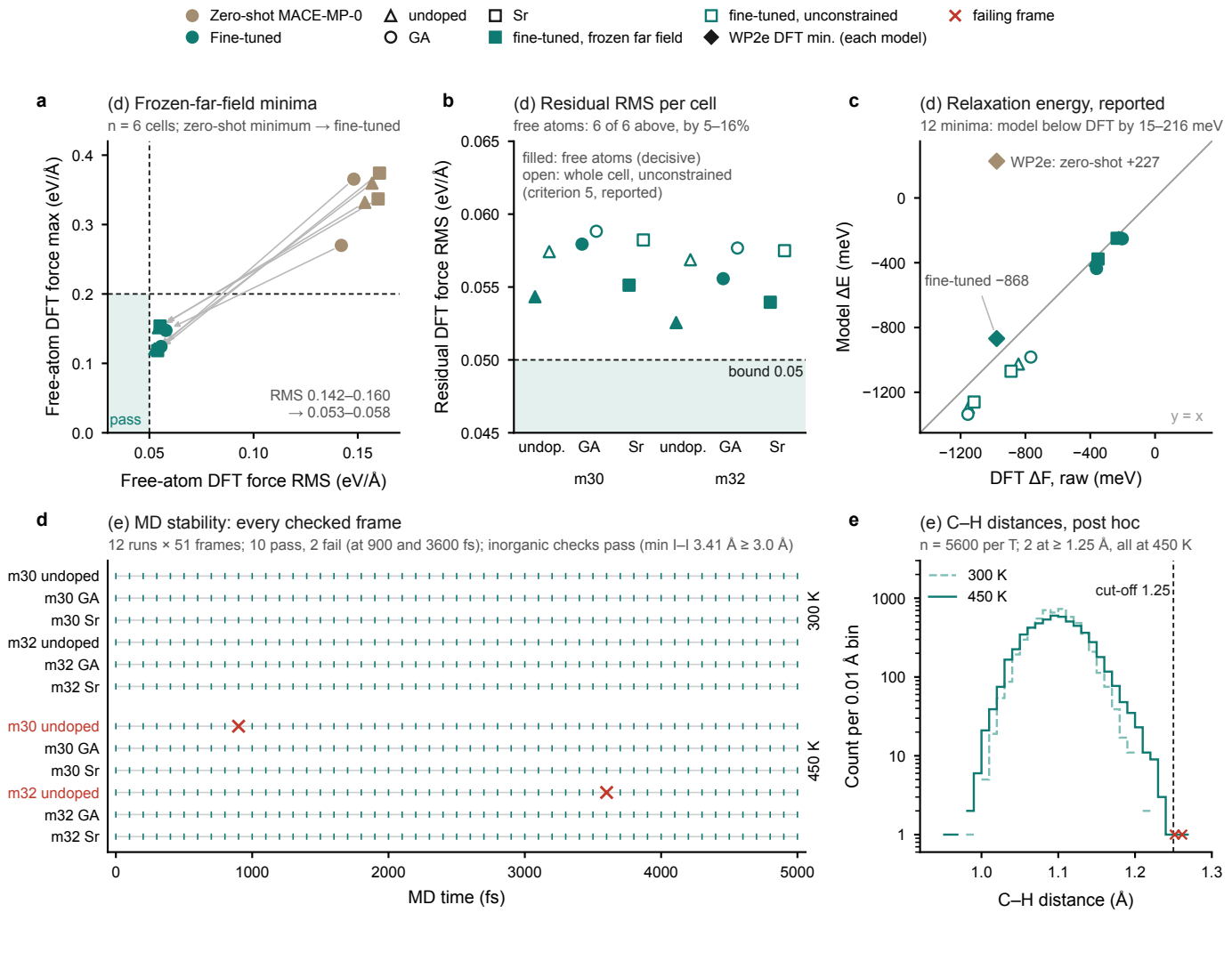

Zero-shot MACE-MP-0
Fine-tuned
undoped
GA
Sr
fine-tuned, frozen far field
fine-tuned, unconstrained
WP2e DFT min. (each model)
failing frame
a
(d) Frozen-far-field minima
n = 6 cells; zero-shot minimum → fine-tuned
Free-atom DFT force max (eV/Å)
Free-atom DFT force RMS (eV/Å)
pass
RMS 0.142–0.160
→ 0.053–0.058
b
(d) Residual RMS per cell
free atoms: 6 of 6 above, by 5–16%
filled: free atoms (decisive)
open: whole cell, unconstrained
(criterion 5, reported)
Residual DFT force RMS (eV/Å)
bound 0.05
undop.
GA
Sr
m30
m32
c
(d) Relaxation energy, reported
12 minima: model below DFT by 15–216 meV
WP2e: zero-shot +227
fine-tuned −868
Model ΔE (meV)
DFT ΔF, raw (meV)
y = x
d
(e) MD stability: every checked frame
12 runs × 51 frames; 10 pass, 2 fail (at 900 and 3600 fs); inorganic checks pass (min I–I 3.41 Å ≥ 3.0 Å)
m30 undoped
m30 GA
m30 Sr
m32 undoped
m32 GA
m32 Sr
300 K
450 K
MD time (fs)
e
(e) C–H distances, post hoc
n = 5600 per T; 2 at ≥ 1.25 Å, all at 450 K
300 K
450 K
cut-off 1.25
Count per 0.01 Å bin
C–H distance (Å)